

12. ALGORITHM OF DEVELOPMENT

DURATION : 10:38

STUDY SHEET

COURSE SUMMARY

This session explores the fundamental algorithm of embryonic development, detailing the transition from oocyte to zygote and the crucial early stages of cell division. The concepts of polarity, asymmetric cell divisions, and increased metabolic potential are discussed, illustrating how a single cell begins to multiply and organise itself. The session also highlights the hatching phase, where accumulated energy reaches a critical threshold, resulting in embryo migration. A fascinating overview of the basic mechanisms underlying embryonic growth, essential for therapeutic practitioners.

12. DEVELOPMENTAL ALGORITHM: FROM OOCYTE TO BLASTOCOEL

For the first three to four days of development, the **oocyte** we observe is no longer an oocyte, but is now called a **zygote**.

It is crucial to understand a **fundamental algorithm** that will constantly repeat throughout embryonic development. This process relies on the concepts of **growth** and **developmental movement** within this growth.

POLARITY AND ASYMMETRICAL CELL DIVISIONS

Here we have a clear image of **polarity**. The first two cells of the zygote will divide extremely rapidly and increasingly to form a cluster of small cells.

These first two living cells respond to a polarity, which means they will not initially divide at the same rate. This difference in division speed is a manifestation of polarity.

With each cell division, the **membrane** increases in surface area, and there is a slight loss of fluid outwards, a small **exudate**. This contributes to the formation of a mass of cells, with the appearance of a first small **fluid-filled cavity** within it.

Initially, during the first few days, the whole multiplies but does not grow in size. Due to cell multiplication and the small loss of water, there is an overall increase in the membrane and cells within a confined space.

This implies an increase in overall **metabolism** and, consequently, an increase in the developing **potential energy**.

Let's recap for a better understanding: initially, we have the first two cells in a confined space, enveloped by a membrane called the **zona pellucida**.

These two cells each have a **nucleus**, never centred, always off-centre. Given cell A and cell B in a plane, depending on the polarity, one cell, closer to a metabolic field, will divide slightly faster than the other.

- **Difference in division speed:** The cell closest to a metabolic field divides faster.
- **Increase in exchange surface:** Each cell division or cleavage leads to an increase in membrane length, thus increasing the exchange surface.
- **Exudate:** A little fluid escapes to the side, forming an exudate.

It is important to note that this division is never symmetrical in the body, which demonstrates that there is no **perfect symmetry**, but rather **harmony**.

THE INCREASE IN METABOLIC POTENTIAL

Initially, we have two cells and two nuclei, offering a certain exchange capacity. Then, there are three cells and three nuclei, which means a greater metabolic exchange force.

The **energetic** and metabolic potential is more or less strong depending on the position in the cell, with more powerful areas. Each division represents a significant transformation and opening.

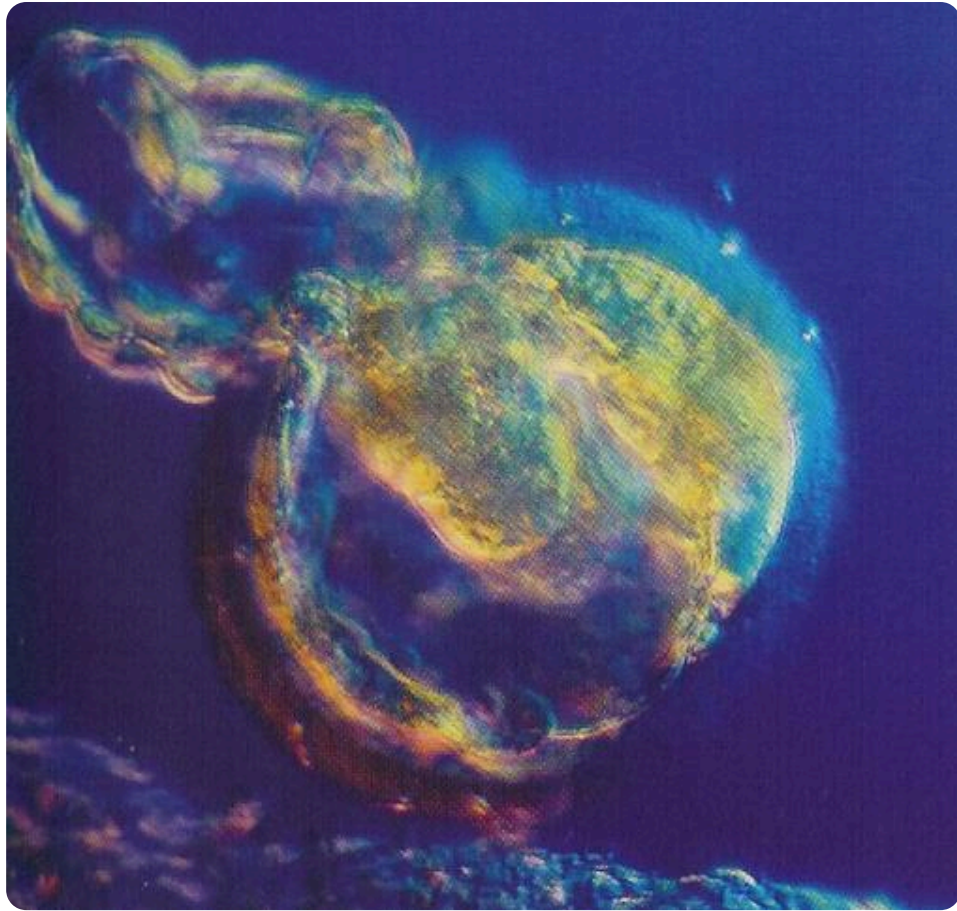


FIG. — Segmenting blastocyst

THE CONCENTRATION PHASE AND HATCHING

During the first three to four days, the whole divides but does not grow, remaining in the same space. It is as if an internal power is concentrating, creating a mass of **metabolic energy** ready to "explode" and seek new trophic forces.

So we have a mass energy, a potential energy that builds up, a very intense metabolic power that concentrates for three days.

This concentration reaches a sufficient level around the 3rd-4th day to cause a phenomenon of **hatching**. The cell, globally enclosed in a tissue, undergoes a rupture due to the accumulated force, allowing the embryo to migrate.

Just before hatching, within this layer, each cell division causes:

- An overall increase in membrane size.
- An increase in the number of cells, and therefore in metabolism, force, and power.
- An exudate.

The very first cavity that appears, filled with water, joins and forms a small cavity.

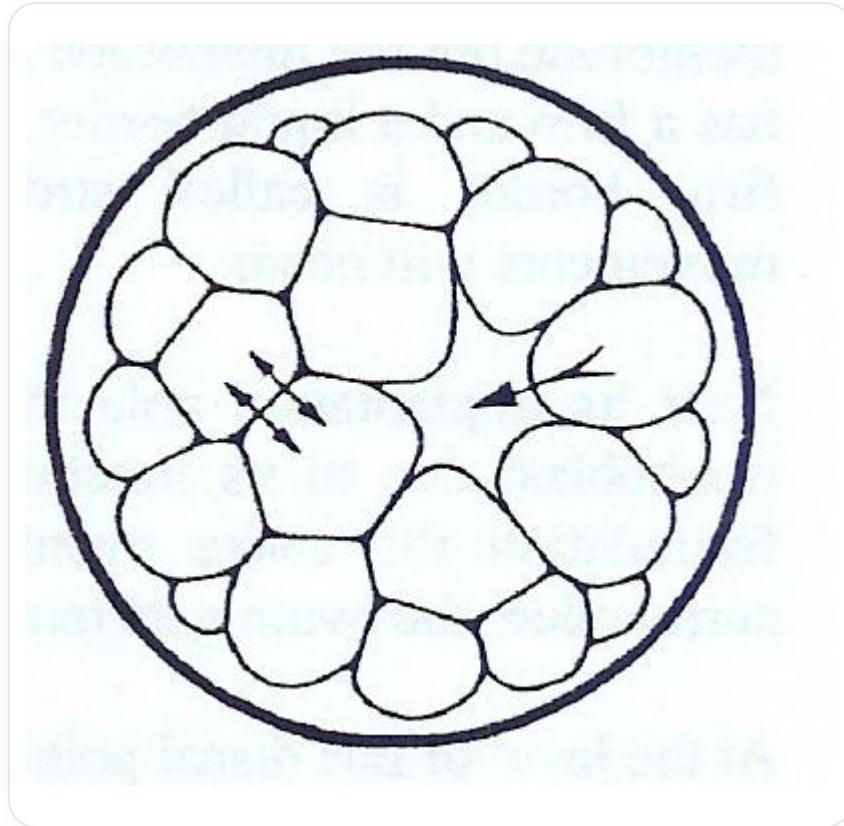


FIG. — Blastocyst cavitation

THE BLASTOCOEL: FIRST FLUID CAVITY

This small cavity is called the **blastocoel**. The term "blastocoel" literally means "germ of the sky", and this cavity will be of great importance for the future **peritoneal cavity**.

The phenomenon of hatching results in a multitude of cells and a cavity. This cavity is always off-centre, the expression of a global polarity.

The eccentricity of this cavity leads to a concentration of cells at one pole, which is called the **animal pole** (as opposed to the vegetal pole), or assimilation pole. At this location, there is a high concentration of cells, less elsewhere.

Between this phase and the next, hatching occurs. The peripheral membrane can no longer contain the whole. A rupture occurs, and the embryo begins to leave this space.

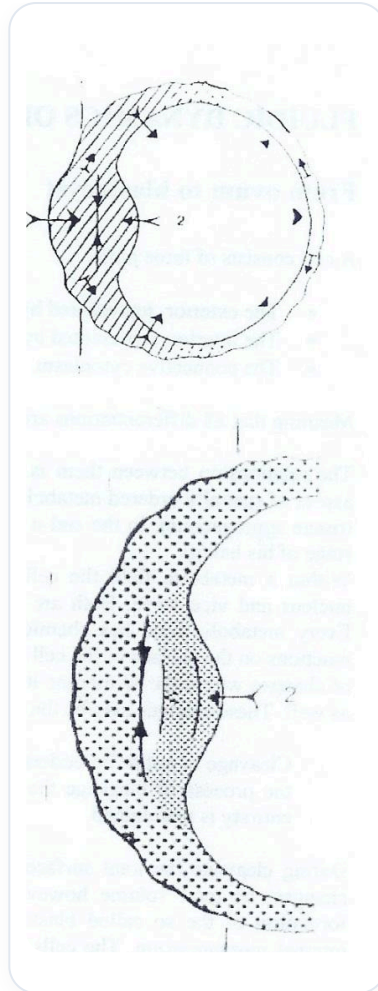


FIG. — Embryonic pinching by contraction

The passage of this first cavity containing the exudate, forming the blastocoel, will grow. The difference from the previous phase is that now, the embryo is much larger because it has emerged from the membrane.

It is crucial to remember that during the first three days, the embryo does not grow. But inside, power accumulates, until the surrounding tissue ruptures and hatching occurs.

The embryo leaves its "egg" and finds itself in the external environment. In this external environment, it very quickly organises itself into a concentrated area of cells and a cavity.

It is still the same image of the blastocoel, but this time, it will develop what is called an **embryonic pole**, with a greater concentration of cells on one side than on the other. This pole is an assimilation pole, seeking to "take in the inside".

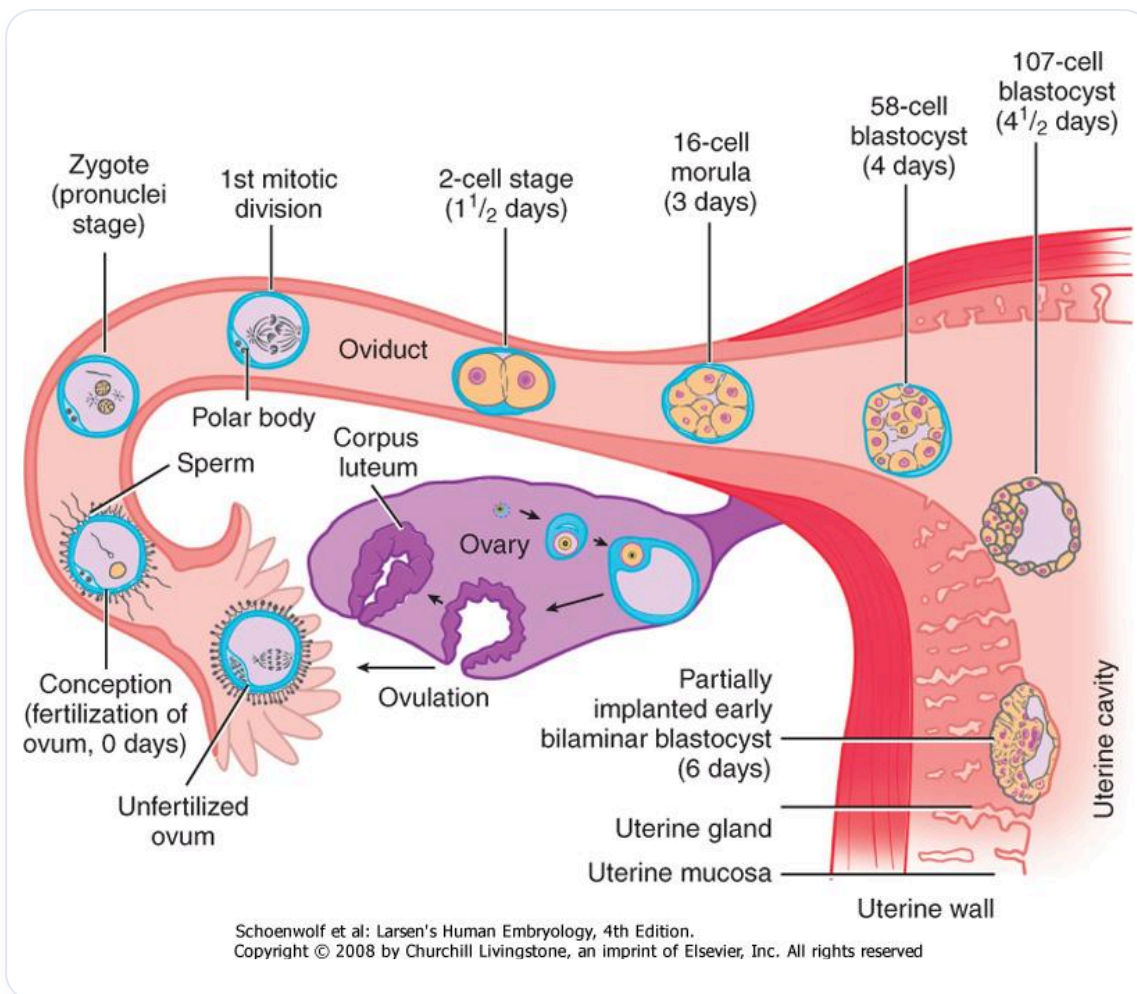


FIG. — Early embryonic development



FIG. — 2-cell embryo

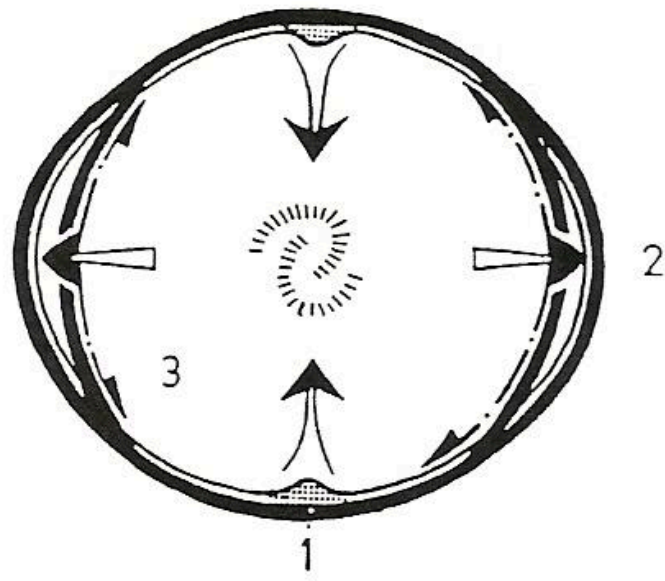
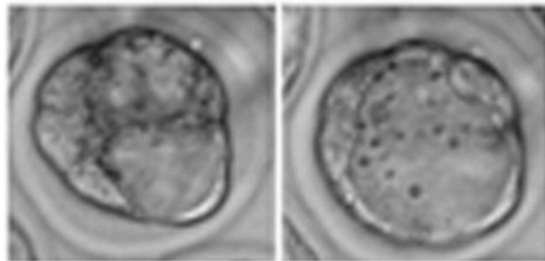


FIG. — *Deformed embryonic shell*

la cavitation

16 cellules



blastocyste

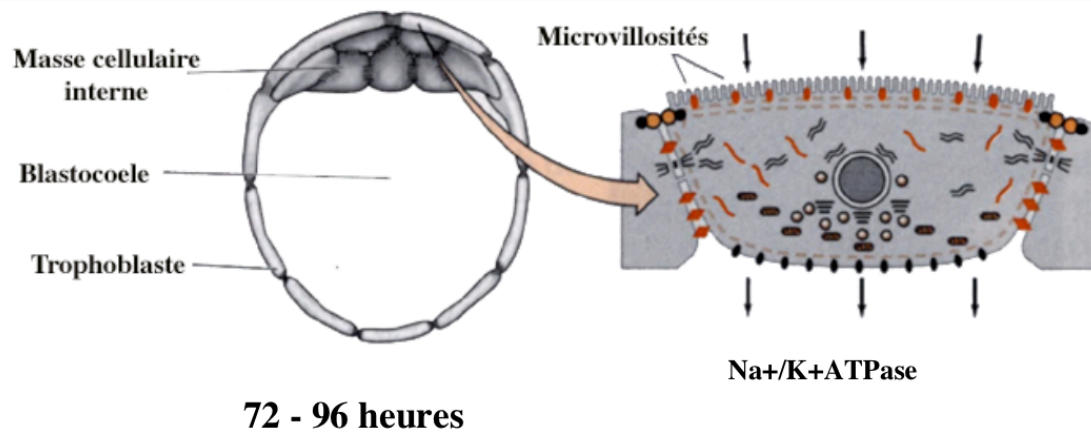
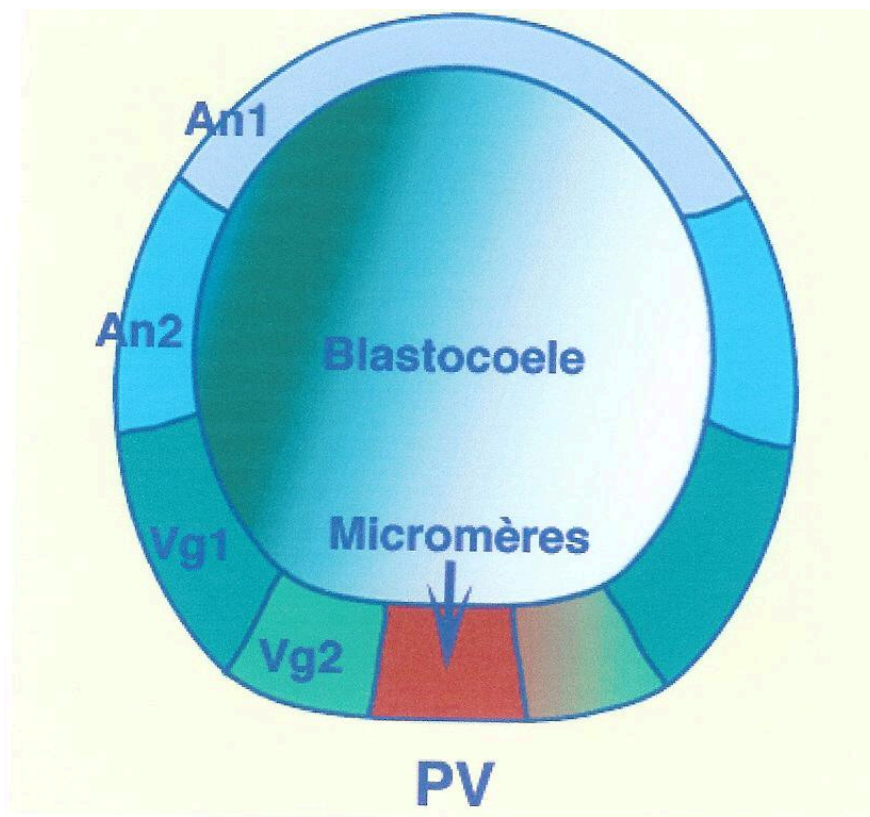


FIG. — Blastocyst cavitation NaKATPase



An1: Animal 1; An2: Animal 2; Vg1: Végétatif 1; Vg2: Végétatif 2; PA: Pôle Animal; PV: Pôle Végétatif.

FIG. — *Blastula Micromeres PV*

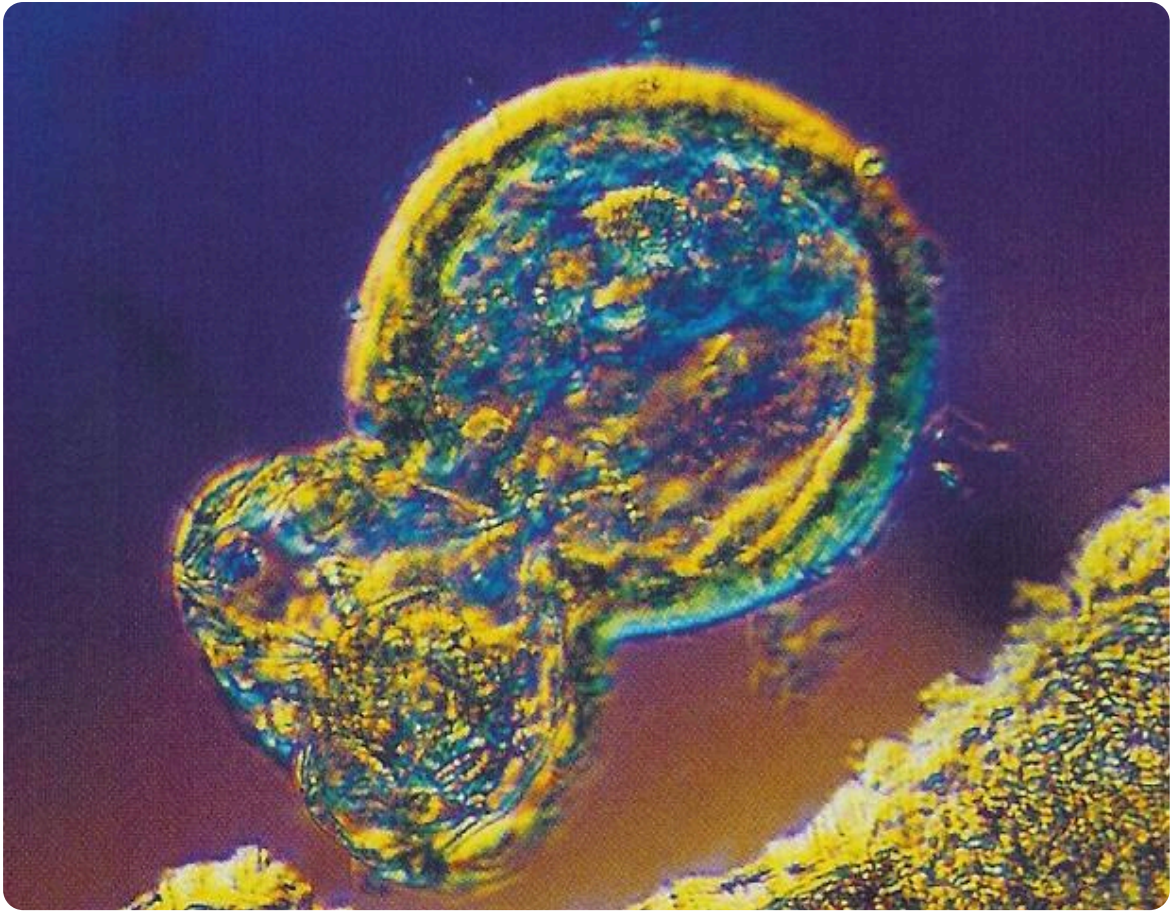


FIG. — Gastropod embryo